

Problem 9.20

A deposit is made on January 1, 2004. The investment earns interest at a rate equivalent to a rate of discount of 6% convertible quarterly. Calculate the monthly effective interest rate for the month of December 2004.

Solution

The nominal annual discount rate convertible quarterly is

$$d^{(4)} = 0.06.$$

Thus, the effective quarterly discount rate is

$$d_q = \frac{0.06}{4} = 0.015.$$

A quarterly discount rate of 0.015 means that the quarterly accumulation factor is

$$\frac{1}{1 - d_q} = \frac{1}{1 - 0.015}.$$

Since each quarter contains three months, let j be the monthly effective interest rate. Then

$$(1 + j)^3 = \frac{1}{1 - 0.015}.$$

Therefore,

$$1 + j = (1 - 0.015)^{-1/3}.$$

Solving for j gives

$$\begin{aligned} j &= (1 - 0.015)^{-1/3} - 1 \\ &\approx 0.005058. \end{aligned}$$

Hence, the monthly effective interest rate for December 2004 is

$$\boxed{0.5058\%}.$$